Problem 5: Zigzag Iterator

Problem Statement

Given two `1d` vectors, implement an iterator to return their elements alternately.

For example, given two `1d` vectors:

```
v1 = [1, 2]
v2 = [3, 4, 5, 6]
```

By calling `next()` repeatedly until `hasNext()` returns `false`, the order of elements returned by next should be: [1, 3, 2, 4, 5, 6].

Example 1:

```
i = ZigzagIterator([1, 2], [3, 4, 5, 6])
print(i.next())  # returns 1
print(i.next())  # returns 3
print(i.next())  # returns 2
print(i.next())  # returns 4
print(i.next())  # returns 5
print(i.next())  # returns 6
print(i.hasNext())  # returns False
```

Example 2:

```
i = ZigzagIterator([1, 2, 3, 4], [5, 6])
print(i.next())  # returns 1
print(i.next())  # returns 5
print(i.next())  # returns 2
print(i.next())  # returns 6
print(i.next())  # returns 3
print(i.next())  # returns 4
print(i.hasNext())  # returns False
```

Example 3:

```
i = ZigzagIterator([1, 2], [])
print(i.next())  # returns 1
print(i.next())  # returns 2
print(i.hasNext())  # returns False
```

Try it yourself

Try solving this question here:

Java

```
import java.util.*;

class Solution {
    public Solution(List<Integer> v1, List<Integer> v2) {
        // ToDo: Write Your Code Here.
    }

    public int next() {
        // ToDo: Write Your Code Here.
        return -1;
    }

    public boolean hasNext() {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

Test Case | Run Code Result | Stdout

```
[1, 2]
[3, 4, 5, 6]
```

Use Example Testcases | Run | Submit

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